NLP PROJECT:
MULTI-LABEL FINANCIAL TEXT CLASSIFICATION ON

Abstract

Multi-label financial text classification is an NLP task that involves assigning one or more topic labels from a fixed taxonomy to short news articles. The Reuters-21578 corpus — a standard benchmark in information retrieval — contains wire-service news dispatches tagged with commodity, economic, and financial categories. The classification challenge is compounded by heavy class imbalance (earn/acq dominate), overlapping vocabulary across related categories (grain/wheat/corn), and ambiguous multi-label assignments.

Dataset

The Reuters-21578 corpus is a widely used benchmark of financial and commodity newswire articles. Each article is associated with one or more topic labels from a vocabulary of over 90 categories. Two preprocessed versions of the dataset were used:

- **df_traditional.csv** — 11,302 articles preprocessed with stemming (Porter stemmer), SGML tag extraction, entity normalization, and stopword removal, implemented in C.
- **df_advanced.csv** — 11,302 articles with lighter preprocessing (no stemming), used for transformer fine-tuning where subword tokenization is preferred.

For all learning-based experiments, the top 10 most frequent labels were retained (earn, acq, money-fx, crude, grain, trade, interest, wheat, ship, corn), yielding a working set of 9,277 labeled documents split 70/10/20 into train/validation/test.

Approach 1: C-Based Preprocessing Pipeline

A complete text normalization pipeline was implemented in C to produce clean, stemmed token sequences suitable for downstream classifiers. The pipeline processes raw Reuters SGML files through the following stages:

- **SGML extraction** — tag-aware parser extracts TITLE and BODY fields
- **Tag cleaning** — residual inline SGML markup is stripped
- **Entity normalization** — dollar amounts → AMOUNT, stock tickers → TICKER, percentages → PERCENT, fiscal quarters → FISCALQUARTER, time ranges → TIMERANGE
- **Lowercasing & tokenization** — alphanumeric tokens extracted with punctuation handling
- **Stopword removal** — domain-specific list of ~80 financial and grammatical stopwords
- **Porter stemming** — lightweight suffix-stripping stemmer (e.g. earnings → earn, acquired → acquir, prices → pric)

The stemmed output formed the basis of all traditional embedding experiments. A sample demonstrating each pipeline stage was verified by trace-printing intermediate token states, confirming correct normalization of financial entities.

Approach 2: Rule-Based Classification

A purely rule-based multi-label classifier was applied directly to the stemmed df_traditional.csv dataset. Each of the 90+ Reuters categories was given a hand-crafted rule dictionary containing:

- **keywords** — stemmed domain terms contributing TF-style scores (e.g. earn: 'earn', 'profit', 'qtr', 'shr', 'dividend')
- **patterns** — regex structural patterns worth 2× keyword weight (e.g. r'\b(net|oper)\s+(profit|loss)\b')
- **neg_kw** — penalizing terms that reduce score by 30% per hit
- **weight** — per-category multiplier to compensate for rare labels

Scoring & Thresholding

Raw scores were normalized by the maximum score across all categories, and any category exceeding a threshold of 0.25 (normalized) was included in the prediction set. At least one label was always predicted by falling back to the highest-scoring category.

Results and Failure Analysis

The rule-based system produced the following overall metrics on the full 11,302-document dataset:

Metric	Value
Exact Match Accuracy	~25–30%
Subset Match Accuracy (any correct label)	~75%
Micro F1	~0.55–0.65 (varies by threshold)
Macro F1	~0.40–0.55

Four failure categories were identified:

Type	Description	Example
A — False Negative	True label entirely missed	'gnp' missed when 'net' keywords fire earn
B — False Positive	Spurious extra labels predicted	'ship' fires on linseed articles due to 'lin' token
C — Mixed (FP+FN)	Some right, some wrong	grain + wheat correct but corn missed
D — Total Wrong	No overlap between true and predicted	money-supply scored as money-fx

Root cause analysis identified five systematic failure patterns: (1) shared stemmed vocabulary where 'earn/net/profit' fired across unrelated categories; (2) rare labels with few distinguishing keywords (dlr, bop, ipi, yen); (3) multi-label co-occurrence between grain/wheat/corn sharing nearly identical term sets; (4) short documents under 30 tokens lacking sufficient scoring context; and (5) stemming collisions such as 'lin' matching both linseed and shipping-line contexts.

Approach 3: Traditional Embeddings + Classical ML Classifiers

Architecture

The stemmed df_traditional.csv corpus was used with a standard multi-label pipeline: feature extraction → OneVsRest classifier. Three sparse and four dense embedding strategies were evaluated against ten classifier algorithms. The dataset was split 70/10/20 (train: 6,493 · val: 928 · test: 1,856). Top-10 labels were retained (9,277 documents).

Sparse Embeddings

- Bag of Words (BoW) — CountVectorizer, max 15,000 features, min_df=2
- TF-IDF (word-level) — sublinear_tf=True, max 15,000 features
- TF-IDF n-gram (1,2) — max 20,000 features, captures bigram patterns

Dense Embeddings

- Word2Vec CBOW — 100d, window=5, trained on train+val corpus
- Word2Vec Skip-gram — 100d, window=5, trained on train+val corpus
- FastText — 100d, subword-aware, trained on train+val corpus
- GloVe — 100d pretrained (6B tokens), document vectors by mean pooling

Classifiers

All models wrapped in OneVsRestClassifier for multi-label support: Logistic Regression, Naive Bayes (ComplementNB), Decision Tree, Random Forest (n=200), SVM (LinearSVC), KNN, Bagging, AdaBoost, Gradient Boosting, XGBoost.

Results — Sparse Embeddings

```
--- BoW ---
  [BoW] LogisticRegression     Micro-F1=0.947 Macro-F1=0.914 (2.6s)
  [BoW] NaiveBayes             Micro-F1=0.869 Macro-F1=0.793 (0.0s)
  [BoW] DecisionTree           Micro-F1=0.874 Macro-F1=0.828 (2.4s)
  [BoW] RandomForest           Micro-F1=0.903 Macro-F1=0.809 (22.2s)
  [BoW] SVM                    Micro-F1=0.943 Macro-F1=0.909 (1.1s)
  [BoW] KNN                    Micro-F1=0.920 Macro-F1=0.858 (3.0s)
  [BoW] Bagging                Micro-F1=0.894 Macro-F1=0.861 (16.2s)
  [BoW] AdaBoost               Micro-F1=0.840 Macro-F1=0.786 (23.4s)
  [BoW] GradientBoosting       Micro-F1=0.911 Macro-F1=0.872 (43.5s)
  [BoW] XGBoost                Micro-F1=0.927 Macro-F1=0.897 (5.0s)

--- TF-IDF ---
  [TF-IDF] LogisticRegression     Micro-F1=0.926 Macro-F1=0.878 (1.2s)
  [TF-IDF] NaiveBayes             Micro-F1=0.833 Macro-F1=0.748 (0.0s)
  [TF-IDF] DecisionTree           Micro-F1=0.879 Macro-F1=0.824 (4.9s)
  [TF-IDF] RandomForest           Micro-F1=0.900 Macro-F1=0.787 (22.2s)
  [TF-IDF] SVM                    Micro-F1=0.953 Macro-F1=0.924 (0.5s)
  [TF-IDF] KNN                    Micro-F1=0.891 Macro-F1=0.845 (2.9s)
  [TF-IDF] Bagging                Micro-F1=0.903 Macro-F1=0.869 (25.5s)
  [TF-IDF] AdaBoost               Micro-F1=0.858 Macro-F1=0.827 (36.3s)
  [TF-IDF] GradientBoosting       Micro-F1=0.911 Macro-F1=0.871 (91.5s)
  [TF-IDF] XGBoost                Micro-F1=0.931 Macro-F1=0.904 (28.1s)

--- TF-IDF-Ngram ---
  [TF-IDF-Ngram] LogisticRegression     Micro-F1=0.929 Macro-F1=0.880 (1.3s)
  [TF-IDF-Ngram] NaiveBayes             Micro-F1=0.849 Macro-F1=0.755 (0.1s)
  [TF-IDF-Ngram] DecisionTree           Micro-F1=0.882 Macro-F1=0.838 (7.1s)
  [TF-IDF-Ngram] RandomForest           Micro-F1=0.911 Macro-F1=0.808 (28.9s)
  [TF-IDF-Ngram] SVM                    Micro-F1=0.953 Macro-F1=0.921 (0.7s)
  [TF-IDF-Ngram] KNN                    Micro-F1=0.914 Macro-F1=0.855 (3.1s)
  [TF-IDF-Ngram] Bagging                Micro-F1=0.903 Macro-F1=0.866 (40.1s)
  [TF-IDF-Ngram] AdaBoost               Micro-F1=0.859 Macro-F1=0.816 (52.5s)
  [TF-IDF-Ngram] GradientBoosting       Micro-F1=0.910 Macro-F1=0.878 (142.2s)
  [TF-IDF-Ngram] XGBoost                Micro-F1=0.933 Macro-F1=0.908 (36.7s)
```

Results — Dense Embeddings

```
--- Word2Vec-CBOW ---
  [Word2Vec-CBOW] LogisticRegression     Micro-F1=0.884 Macro-F1=0.792 (0.3s)
  [Word2Vec-CBOW] DecisionTree           Micro-F1=0.881 Macro-F1=0.787 (7.1s)
  [Word2Vec-CBOW] RandomForest           Micro-F1=0.919 Macro-F1=0.823 (32.0s)
  [Word2Vec-CBOW] SVM                    Micro-F1=0.895 Macro-F1=0.809 (0.9s)
  [Word2Vec-CBOW] KNN                    Micro-F1=0.921 Macro-F1=0.846 (2.3s)
  [Word2Vec-CBOW] Bagging                Micro-F1=0.902 Macro-F1=0.800 (35.0s)
  [Word2Vec-CBOW] AdaBoost               Micro-F1=0.905 Macro-F1=0.792 (53.7s)
  [Word2Vec-CBOW] GradientBoosting       Micro-F1=0.910 Macro-F1=0.810 (192.7s)
  [Word2Vec-CBOW] XGBoost                Micro-F1=0.932 Macro-F1=0.857 (4.6s)

--- Word2Vec-SG ---
  [Word2Vec-SG] LogisticRegression     Micro-F1=0.894 Macro-F1=0.813 (0.2s)
  [Word2Vec-SG] DecisionTree           Micro-F1=0.854 Macro-F1=0.766 (7.5s)
  [Word2Vec-SG] RandomForest           Micro-F1=0.913 Macro-F1=0.820 (33.2s)
  [Word2Vec-SG] SVM                    Micro-F1=0.908 Macro-F1=0.832 (0.6s)
  [Word2Vec-SG] KNN                    Micro-F1=0.928 Macro-F1=0.862 (2.1s)
  [Word2Vec-SG] Bagging                Micro-F1=0.890 Macro-F1=0.773 (39.1s)
  [Word2Vec-SG] AdaBoost               Micro-F1=0.904 Macro-F1=0.811 (54.7s)
  [Word2Vec-SG] GradientBoosting       Micro-F1=0.910 Macro-F1=0.810 (197.3s)
  [Word2Vec-SG] XGBoost                Micro-F1=0.934 Macro-F1=0.867 (5.1s)


--- FastText ---
  [FastText] LogisticRegression     Micro-F1=0.876 Macro-F1=0.778 (0.4s)
  [FastText] DecisionTree           Micro-F1=0.883 Macro-F1=0.794 (6.5s)
  [FastText] RandomForest           Micro-F1=0.916 Macro-F1=0.817 (31.4s)
  [FastText] SVM                    Micro-F1=0.881 Macro-F1=0.784 (1.0s)
  [FastText] KNN                    Micro-F1=0.921 Macro-F1=0.837 (2.1s)
  [FastText] Bagging                Micro-F1=0.901 Macro-F1=0.799 (32.4s)
  [FastText] AdaBoost               Micro-F1=0.898 Macro-F1=0.794 (54.7s)
  [FastText] GradientBoosting       Micro-F1=0.911 Macro-F1=0.806 (196.1s)
  [FastText] XGBoost                Micro-F1=0.930 Macro-F1=0.857 (4.7s)

--- GloVe ---
  [GloVe] LogisticRegression     Micro-F1=0.831 Macro-F1=0.739 (0.3s)
  [GloVe] DecisionTree           Micro-F1=0.768 Macro-F1=0.683 (8.3s)
  [GloVe] RandomForest           Micro-F1=0.839 Macro-F1=0.705 (37.8s)
  [GloVe] SVM                    Micro-F1=0.829 Macro-F1=0.736 (0.7s)
  [GloVe] KNN                    Micro-F1=0.897 Macro-F1=0.839 (2.1s)
  [GloVe] Bagging                Micro-F1=0.822 Macro-F1=0.667 (44.6s)
  [GloVe] AdaBoost               Micro-F1=0.837 Macro-F1=0.720 (56.2s)
  [GloVe] GradientBoosting       Micro-F1=0.857 Macro-F1=0.748 (196.2s)
  [GloVe] XGBoost                Micro-F1=0.882 Macro-F1=0.797 (5.8s)
```

Note: GloVe embeddings produced lower silhouette scores (0.095) compared to Word2Vec-CBOW (0.268) and Word2Vec-SG (0.213), indicating poorer cluster separability on this domain-specific corpus.

Best Overall Combination

Embedding	Model	Accuracy	Micro F1	Macro F1
TF-IDF-Ngram (1,2)	SVM (LinearSVC)	0.9111	0.9534	0.9212

Per-Label Breakdown (Best Model: TF-IDF-Ngram + SVM)

Label	Precision	Recall	F1	Support
acq	0.96	0.99	0.97	492
corn	0.88	0.92	0.90	49
crude	0.90	0.94	0.92	126
earn	0.99	0.98	0.98	792
grain	0.93	0.93	0.93	124
interest	0.82	0.94	0.88	98
money-fx	0.86	0.93	0.90	163
ship	1.00	0.82	0.90	61
trade	0.92	0.98	0.95	96
wheat	0.86	0.90	0.88	61
micro avg	0.95	0.96	0.95	2062
macro avg	0.91	0.93	0.92	2062

Analysis: Why SVM + TF-IDF outperforms dense embeddings here

TF-IDF provides high-dimensional, term-discriminative features ideal for financial newswire, where distinct vocabulary (e.g. 'dlrs', 'qtr', 'crude') strongly indexes categories. LinearSVC finds maximum-margin hyperplanes in this sparse space efficiently. Dense embeddings (Word2Vec, FastText) sacrifice discriminability for generalization: semantically similar but categorically distinct words collapse into nearby vectors, hurting precision on overlapping financial categories. GloVe performed worst among dense methods as its global co-occurrence statistics were trained on general-domain text, while Reuters vocabulary is highly domain-specific.

Silhouette analysis confirmed this: Word2Vec-CBOW achieved the highest embedding separability (0.268) but still underperformed TF-IDF-based models in classification, suggesting that geometric cluster quality does not directly translate to classification accuracy for sparse, high-cardinality label sets.

Hyperparameter Tuning

Grid search (3-fold CV) was performed on the top two combinations:

- TF-IDF + SVM: best C=1.0 → Micro F1=0.9528 (no change, already optimal)
- TF-IDF + Logistic Regression: best C=5.0, solver=lbfgs → Micro F1=0.9447 (+0.02)

Approach 4: Transformer Fine-Tuning

Three transformer-based approaches were explored on the non-stemmed df_advanced.csv corpus, where subword tokenization benefits from preserved surface forms. All models used the top-10 label set with the same 80/20 train/val split. Multi-label classification was formulated as sigmoid outputs with Binary Cross-Entropy loss, and predictions were made by thresholding the output probabilities.

4a. RoBERTa-base (Fine-tuned, Layers 10–11 + Classifier)

roberta-base was fine-tuned for 5 epochs (lr=2e-5, batch=8, max_length=256). Lower layers were frozen; only the top two transformer layers and the classification head were trained. Training loss converged to 0.0494.

```
Threshold 0.3: {'accuracy': 0.932620320855615, 'f1_micro': 0.9647283126787417, 'f1_macro': 0.9374894140190315}
Threshold 0.4: {'accuracy': 0.9390374331550803, 'f1_micro': 0.9675870348139256, 'f1_macro': 0.9389101275876616}
Threshold 0.5: {'accuracy': 0.9427807486631016, 'f1_micro': 0.968273189634294, 'f1_macro': 0.9405988572083969}
Threshold 0.6: {'accuracy': 0.9449197860962567, 'f1_micro': 0.96923828125, 'f1_macro': 0.9392962818500219}
```

Optimal threshold was 0.5–0.6. RoBERTa-base achieved the best results across all transformer configurations, with Micro F1 of 0.969 at threshold 0.6.

4b. DistilBERT-base-uncased (Fine-tuned, Layers 10–11 + Classifier)

distilbert-base-uncased was fine-tuned under identical settings. Training loss converged to 0.0682 (higher than RoBERTa), reflecting DistilBERT's reduced capacity from knowledge distillation.

```
Threshold 0.3: {'accuracy': 0.9037433155080213, 'f1_micro': 0.9397937155193092, 'f1_macro': 0.8935463967264029}
Threshold 0.4: {'accuracy': 0.9080213903743316, 'f1_micro': 0.9417733752114037, 'f1_macro': 0.8976929312753568}
Threshold 0.5: {'accuracy': 0.9106951871657754, 'f1_micro': 0.9423778264040846, 'f1_macro': 0.8971833988141021}
Threshold 0.6: {'accuracy': 0.9101604278074866, 'f1_micro': 0.9416952474277315, 'f1_macro': 0.896098717568442}
```

DistilBERT achieved Micro F1 of ~0.942, around 2.7 points below RoBERTa, consistent with its smaller architecture (~40% fewer parameters than BERT-base).

4c. BERT-base + SVM Classifier (Frozen Encoder)

BERT-base-uncased was used purely as a frozen feature extractor. CLS token embeddings (768d) were extracted for all training and validation documents, then a OneVsRest LinearSVC was trained on these representations.

```
F1 Micro: 0.8911368015414258
F1 Macro: 0.8380346280456863
```

The frozen BERT encoder underperformed both fine-tuned models. Without task-specific fine-tuning, the CLS token captures generic sentence semantics rather than the category-discriminative patterns needed for financial topic classification. LinearSVC over frozen BERT (0.891 Micro F1) is also weaker than TF-IDF + SVM (0.953), reinforcing that dense general-purpose encoders need fine-tuning to match sparse TF-IDF on domain-specific classification.

Comparison of All Approaches

Approach	Best Configuration	Micro F1	Macro F1	Notes
Rule-Based	Keyword + regex rules	~0.60	~0.48	No learning; fragile vocabulary
Traditional Embedding + ML	TF-IDF-Ngram + SVM	0.9534	0.9212	Best sparse approach
Dense Embedding + ML	Word2Vec-SG + XGBoost	0.9340	0.8673	Lower than TF-IDF
RoBERTa (fine-tuned)	Top-2 layers + head	0.9692	0.9393	Best overall
DistilBERT (fine-tuned)	Top-2 layers + head	0.9424	0.8972	Fast, moderate perf.
BERT + SVM (frozen)	CLS embed + LinearSVC	0.8911	0.8380	Needs fine-tuning

RoBERTa fine-tuned on the upper two transformer layers achieves the highest Micro F1 (0.969). Notably, TF-IDF-Ngram + SVM is remarkably competitive at 0.953 — within 1.6 points of RoBERTa while being orders of magnitude faster to train. This is characteristic of financial newswire: the vocabulary is highly specialized, making n-gram frequency a near-sufficient signal. The rule-based approach's ~0.60 Micro F1 confirms that hand-crafted heuristics alone, especially on stemmed text without domain expert input, are insufficient for reliable multi-label classification.

Conclusions and Future Work

This project compared four families of NLP pipeline for multi-label financial text classification on Reuters-21578. The main findings are:

- Sparse TF-IDF with SVM remains highly competitive on domain-specific financial text, achieving 0.953 Micro F1 with minimal compute and no training data beyond term statistics.
- Dense embeddings (Word2Vec, FastText) trained in-domain offered no advantage over TF-IDF, while pre-trained GloVe underperformed due to domain mismatch.

- Fine-tuned RoBERTa surpassed all methods at 0.969 Micro F1, demonstrating the value of contextual representations even when only the upper layers are trained.
- Frozen BERT as a feature extractor yields poor results — transformer models must be fine-tuned to be useful on specialized downstream tasks.
- Rule-based systems suffer fundamentally from vocabulary ambiguity in stemmed financial text; expert-curated rules (e.g. m1/m2/m3 for money-supply, boj/plaza for forex) could partially address this but cannot close the gap with learned representations.

Recommended Future Directions

- Fine-tune FinBERT or a Reuters-pretrained model for further gains on financial categories.
- Address class imbalance (earn: 3,902 vs corn: 253) with label-weighted loss or oversampling.
- Explore label-aware attention mechanisms or label embedding classifiers for the multi-label setting.
- Develop a domain-knowledge-based rule set and integrate it as a post-processing step on top of the fine-tuned transformer to handle edge cases the model consistently misclassifies.
- Extend evaluation to the full 90+ Reuters label taxonomy beyond the top-10 subset.